

18/09/2024 (Shift - II)

Total no. of Pages:

Roll no.

Degree: B. TECH. Semester: 3rd — Course work
MID-SEMESTER EXAMINATION, September, 2024

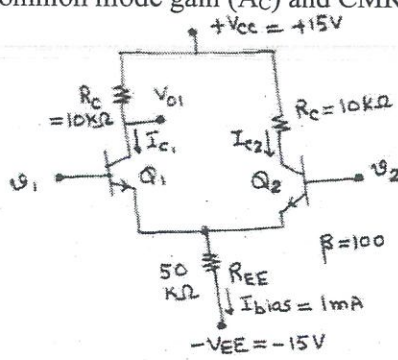
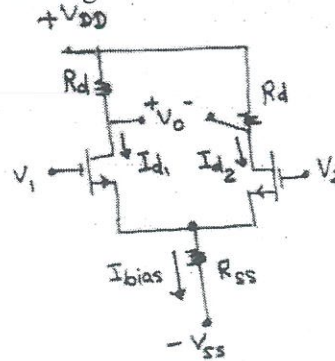
Course Title: Microelectronics Circuits and Applications

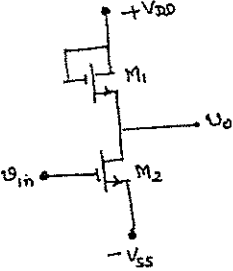
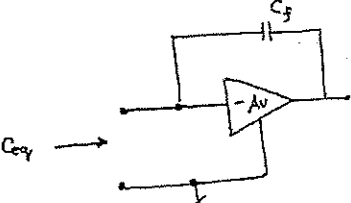
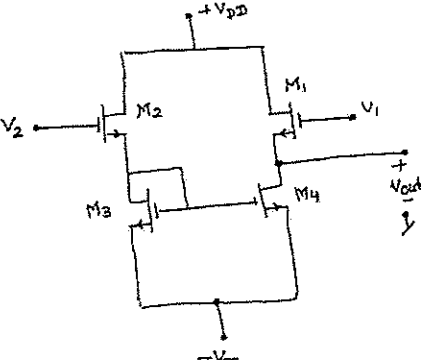
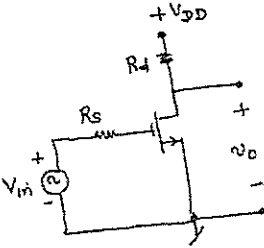
Course Code: EIECC304

Duration: 1:30 Hours

Max. Marks: 20

Note: - Attempt all questions in the given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer

Q. No.	Question	Marks	CO
1a	Explain small signal model of a BJT and define the parameters r_{π} , r_e , g_m and r_o of the BJT.	2	1
1b	Explain the high frequency model of a MOSFET including relevant parasitic capacitances.	2	2
2a	Consider the bipolar differential amplifier circuits given in Fig.1. Determine differential gain (A_d), common mode gain (A_C) and CMRR.  Fig. 1	2	1
2b	By appropriate analysis, determine the linear range of operation for the MOS-differential amplifier shown in Fig.2.  Fig. 2	2	1
3a	Derive an expression for the equivalent small signal resistance of a diode connected MOSFET, also draw its V-I characteristics and explain its role as an active load by giving an example.	2	2

3b	Consider the MOSFET amplifier shown in Fig.3. Assume both the MOSFETs to be operating in saturation. Derive an expression for output voltage (V_o) as a function of V_{in}, V_{ss}, V_{DD} and V_{th} and hence, determine the DC offset voltage present at output.  Fig.3	2	2
4a	Explain Millers theorem and its dual.	2	1
4b	Apply Millers theorem to determine the value of equivalent input capacitance (C_{eq}) of the circuit given in Fig.4.  Fig. 4	2	1
5a	Analyse the differential to single ended converter shown in Fig.5 using square law and hence, determine expression for the AC and DC signal components present in the output of the circuit.  Fig 5	2	1
5b	For the common source amplifier shown in Fig.6, determine an expression for the 3-dB bandwidth of the circuit taking into account the various parasitic capacitances and the parameters of MOSFET.  Fig.6	2	2